Alex McPherson Uribe

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HIGHLIGHTS OF SKILLS

- Co-op student specializing in low-level systems programming and software development; adept at solving complex problems through teamwork and a methodical, process-driven approach.
- Proficient in C, C++, Rust, and Python; experienced with QA, testing, and debugging.
- Strong foundation in data structures, algorithms, software design, mathematical modelling, and networking and security fundamentals.
- Experienced in Linux (Debian/Ubuntu, Fedora); skilled with CLI tooling, Bash, and Git.
- Skilled in building server-side web apps with MySQL, REST APIs, and React/TypeScript frontends.

EDUCATION

Bachelor of Science (Honours), Computer Science Co-op *September 2025 – Present*
Trent University, Peterborough, ON – GPA: 88.0%

- Dean's Honour Roll (GPA 88%) *2025 – Present*
- **Relevant Courses:** Data Structures and Algorithms, Software Design and Modelling, Web Development II: Server-Side, Database Management Systems, Introduction to Data Science

TECHNICAL SKILLS

Languages: C, C++, Python, Java, Rust, C#, JavaScript, TypeScript, PHP
Tools & Frameworks: Git, GitHub, GitHub Actions, React, Bash, CMake, Make, Cargo, SSH, LaTeX
Databases: MySQL, MariaDB, Snowflake
Operating Systems: Linux (Debian/Ubuntu, Fedora), Windows, WSL2

PROJECTS

Fretbored – Music Theory Engine *May 2026 – Present*
Rust

- Designed a high-performance music theory engine in Rust supporting 1,296+ unique chord types across any string instrument tuning, generating 260+ voicings per chord in 7.6 ms.
- Used Rust's ownership model to implement memory-safe data structures and algorithmic optimizations to ensure deterministic execution and eliminate runtime errors.

AMUVIE – Full-Stack Movie Logging Application *June – July 2026*
PHP, React, MySQL, TypeScript, HTML/CSS

- Built a decoupled full-stack app with a PHP/MySQL REST API backend and a React/TypeScript frontend.
- Implemented CRUD functionality across 8 relational database tables with 4 optimized indexes, enabling efficient querying for adding, updating, viewing, and deleting movies.
- Integrated user authentication and authorization across 18 API endpoint operations and 8 React routes (including protected routes), ensuring secure access control.

Trent Marketplace AI Listing Creator *November 2025*
Python, React, Ollama, Snowflake

- Collaborated within a team during a fast-paced hackathon to build a localized AI-powered application using the Ollama framework for automated marketplace listings.
- Integrated Snowflake for scalable backend data retrieval and listing metadata management.
- Tested and debugged external data model integration with local architectures for reliable output.

WORK EXPERIENCE

Guitar & Music Instructor | Self-Employed, Waterloo, ON *September 2022 – Present*

- Designed a structured curriculum teaching guitar and music theory from the fundamentals up.
- Managed scheduling, communication, and lesson planning independently using a custom Notion workflow.
- Retained a student for nearly four years by tracking progress and giving constructive feedback.
- Explained complex music theory concepts technically and concisely under time constraints.